

# Session 09 (AIML) - Assignment Questions

## Q1. (Creating Dictionaries)

Write a program to create dictionaries using different methods:

- a) An empty dictionary using two different ways.
- b) A dictionary with string keys (name, city, course).
- c) A dictionary with integer keys.
- d) A mixed data type dictionary (string, int, float).

Print each dictionary and its type. Add comments explaining the syntax.

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## Q2. (Accessing and Modifying Elements)

Create the dictionary:

```
student = {  
    "name": "YourName",  
    "age": 00,  
    "city": "YourCity",  
    "marks": 00  
}
```

- Print the value of "name" using square brackets.
  - Update the "marks" to 92.
  - Add a new key "grade" with value "A".
  - Print the updated dictionary.
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## Q3. (get(), keys(), values(), items())

Create a dictionary:

```
person = {"name": "Priya", "age": 21, "profession": "Engineer"}
```

Demonstrate:

- a) Using get() to access "age" and a non-existing key "salary" (with default value).
- b) Print all keys using keys().

- c) Print all values using values().
  - d) Print all key-value pairs using items().
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#### Q4. (Nested Dictionary)

Create a nested dictionary to store information of two students:

Python

```
students = {  
    "s1": {"name": "Rahul", "age": 20, "marks": 88},  
    "s2": {"name": "Sneha", "age": 21, "marks": 95}  
}
```

- Print the details of the first student.
  - Print the marks of the second student.
  - Add a new subject "math" with marks 90 for the first student.
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#### Q5. (update() and pop())

Create a dictionary: info = {"brand": "Samsung", "model": "A52", "price": 25000}

- Update it with new information: {"color": "Black", "price": 24000}.
  - Remove the key "model" using pop() and print the removed value.
  - Print the final dictionary.
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#### Q6. (popitem() and clear())

Create a dictionary with at least 5 key-value pairs.

- Use popitem() twice and print the removed items each time.
- Clear the entire dictionary using clear().
- Print the final result.

Add comments explaining the difference between pop() and popitem().

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#### Q7. (copy() and setdefault())

Create a dictionary: d = {"a": 1, "b": 2}

- Make a copy of it.

- Use `setdefault()` to add key "c" with value 3 (if it doesn't exist).
  - Use `setdefault()` again for an existing key "a".  
Print the original and copied dictionaries.
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#### Q8. (fromkeys() and Membership)

Write a program that:

- Uses `dict.fromkeys()` to create a dictionary with keys ["name", "age", "city"] and default value `None`.
  - Takes user input to fill these keys.
  - Checks if a given key exists using `in` operator.
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#### Q9. (Practical Application)

Write a program to store and manage contact information using a dictionary:

- Take name, phone number, and email as input.
  - Store them in a dictionary with name as key.
  - Allow the user to search a contact by name using `get()`.
  - Print all contacts using `items()`.
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#### Q10. (Mini Project - Combined Concepts)

Create a **Student Management System** using a dictionary:

- Repeatedly ask the user to enter student details (roll number as key, and a nested dictionary containing name, age, and marks).
- Use a while loop with a menu:
  1. Add new student
  2. Update marks of a student
  3. Search student by roll number
  4. Display all students
  5. Remove a student
  6. Exit

Use appropriate dictionary methods (`get()`, `update()`, `pop()`, etc.). Add meaningful comments.